

A Frozen Specification for the Measurement of Hardy-Z Breadth in Consecutive Zero Cells

Protocol v2 — 10,000 Hardy Cells
Measurement completed: August 2026

Abstract

This document specifies the complete frozen measurement protocol for a 10,000-cell Hardy-Z breadth survey conducted on consecutive nontrivial zeros of the Riemann zeta function. The protocol employs a dual-resolution architecture (64-point at 28 decimal places precision; 128-point at 40 decimal places precision) to measure the interior maximum of $|Z(t)|$ within each cell, with derivative-free golden-section refinement. Five explicit certification gates ensure numerical stability and data integrity. No hypothesis testing or interpretation was performed during acquisition. All 10,000 cells achieved full certification (all five gates PASS). Refinement discrepancies remained well below preregistered tolerances ($\epsilon_W < 10^{-9}$, $\epsilon_X < 10^{-5}$). This specification is presented to enable independent reproduction of the measurement and to establish an authoritative record of the numerics, thresholds, and procedural safeguards that protected the dataset.

1. Introduction

The distribution of nontrivial zeros of the Riemann zeta function along the critical line remains one of mathematics' most persistent open questions. While large-scale statistical properties of these zeros are well-studied, precision measurement of local geometric properties within zero-separated cells has remained less formalized. This paper documents a complete measurement protocol for extracting a simple but well-defined observable: the width and location of the maximum absolute value of Hardy's Z-function between consecutive pairs of zeros.

The measurement is intentionally limited in scope. It does not attempt to extract or interpret stationary-point information, physical mechanisms, asymptotic behavior, or correlational structure. Instead, it provides a clean, reproducible dataset of interior maxima and their locations under fixed numerical conditions. This narrowness is deliberate: it allows the data to be useful independent of later interpretations and protects against confirmation bias during acquisition.

2. Methods

2.1 Cell Definition

Let γ_n and γ_{n+1} denote consecutive nontrivial zeros of $\zeta(s)$ on the critical line, ordered by ordinate. Define the Hardy cell as the open interval $C_n = (\gamma_n, \gamma_{n+1})$ and the cell width as $h_n = \gamma_{n+1} - \gamma_n$.

2.2 Primary Observable: Transverse Breadth

For each cell C_n , the primary observable is the maximum absolute value of Hardy's Z-function within the cell:

$$M_n = \max_{\{\gamma_n < t < \gamma_{n+1}\}} |Z(t)|$$

The frozen breadth observable is defined as:

$$W_n^\perp = 2 \cdot M_n$$

The factor of two is part of the definition and is not refitted or adjusted from data.

2.3 Maximum Location (Dimensionless Coordinate)

Let $t_{\{\max,n\}}$ denote the location of the selected interior maximum. The dimensionless cell coordinate is:

$$x_n = (t_{\{\max,n\}} - \gamma_n) / (\gamma_{n+1} - \gamma_n)$$

This removes absolute cell width from the location: $x = 0$ corresponds to the lower zero boundary, $x = 1$ to the upper boundary, and $x = 0.5$ to the geometric midpoint.

2.4 Derived Quantity: Breadth-to-Width Ratio

$$\rho_n = W_n^\perp / h_n$$

This is a derived measurement recording the ratio of breadth to cell width.

2.5 Two-Resolution Architecture

Each cell was measured independently at two numerical resolutions to verify convergence:

Resolution A: 64 uniform grid points, 28 decimal places of precision (mpmath dps)

Resolution B: 128 uniform grid points, 40 decimal places of precision

The final reported value used Resolution B (high-resolution); Resolution A served as a convergence check.

2.6 Breadth Search: Derivative-Free Maximization

To avoid importing derivative approximation error into the breadth observable, the maximization was purely value-based:

1. Grid-based scanning: For grid values $y_i = |Z(t_i)|$, identify candidates at interior grid points i where $y_i \geq y_{i-1}$ and $y_i \geq y_{i+1}$.
2. Bracket extraction: Each candidate i provides a local bracket $[t_{i-1}, t_{i+1}]$.
3. Golden-section refinement: Apply 32 iterations of golden-section search to maximize $|Z(t)|$ within each bracket.
4. Selection: If multiple candidates exist, the one with the largest refined $|Z|$ value determines the cell breadth. No $Z'(t)$ information was used.

2.7 Convergence Thresholds (Hard Limits)

Define convergence discrepancies:

$$\varepsilon_W = |W_B - W_A| / \max(|W_A|, |W_B|)$$

$$\varepsilon_x = |x_B - x_A|$$

A cell passes refinement stability if and only if both conditions hold:

$$\varepsilon_W < 10^{-9} \text{ (strictly less-than)}$$

$$\varepsilon_x < 10^{-5} \text{ (strictly less-than)}$$

These thresholds were frozen before any cells were measured and were never modified based on observed results.

2.8 Five-Gate Certification

Each cell underwent five explicit certification gates:

Gate 1 — Breadth Solver: Both A and B breadth calculations must return positive breadths ($W_A > 0$, $W_B > 0$).

Gate 2 — Interior Maximum: Both maxima must lie strictly in the interior: $\gamma_n < t_A < \gamma_{n+1}$ and $\gamma_n < t_B < \gamma_{n+1}$.

Gate 3 — Diagnostic Transparency: Stationary-root diagnostic status must be explicitly recorded. The status UNRESOLVED is permitted; stationary-root success is not required for breadth certification.

Gate 4 — Refinement Stability: Both convergence thresholds must be satisfied: $\varepsilon_W < 10^{-9}$ and $\varepsilon_x < 10^{-5}$.

Gate 5 — Raw Integrity: The measurement must originate from the frozen pipeline: correct zero source, correct cell, no smoothing, no interpolation, no repair of values, no protocol changes.

A cell received status `all_five_gates = PASS` only when every gate passed.

2.9 Zero Source and Reproducibility

The formal run used externally supplied LMFDB zero ordinates from file 10010.txt. A provenance safeguard was strict: no silent substitution of alternative zero generators. The SHA-256 hash of the zero file is:

36ed073e9f5cc5a08eaeac2ca34f8689dd58dbe4ea2fd06e539a20ce4cb06144

To reproduce this measurement, establish byte-for-byte equivalence with this source or use the same LMFDB file.

2.10 Hypothesis Seal

During data acquisition, the following analyses were explicitly forbidden to prevent confirmation bias and hypothesis-driven measurement modification:

- Analysis of x_n distributions
- Compression/decompression language or interpretations
- "Breathing" or cyclical pattern language
- Correlations between breadth and width
- Height evolution language
- Predictive or causal relationships
- Mining for local/global geometric patterns or ratios

This firewall ensured that the dataset captured what was measured, not what patterns the experimenter expected to find.

3. Results

The measurement protocol was applied to all 10,000 consecutive Hardy cells (cells 1 through 10,000). All 10,000 cells achieved full certification (`all_five_gates = PASS`).

Statistic	ϵ_W (Breadth Discrepancy)	ϵ_x (Location Discrepancy)
Mean	4.91×10^{-18}	6.69×10^{-10}
Maximum	2.63×10^{-15}	8.96×10^{-8}
Preregistered Limit	10^{-9}	10^{-5}
Status	PASS (max < limit)	PASS (max < limit)

Table 1: Refinement error statistics across all 10,000 cells. Both mean and maximum discrepancies remained far below preregistered hard limits, confirming numerical stability.

The complete dataset of 10,000 cells is preserved in the master artifact:

breathing_v2_cells_1_10000_master_CERTIFIED.csv

This file contains 10,000 data rows plus header, with 33 fields per cell documenting all A and B measurements, errors, gate statuses, and diagnostic information.

4. Discussion

Scope and Limitations: This protocol measures a single, well-defined observable: the magnitude and location of $|Z(t)|$'s interior maximum within each zero-separated cell. It does not attempt to extract or validate:

- Stationary-point locations ($Z'(t_{\max}) = 0$)
- Asymptotic behavior or scaling laws
- Correlation structures between cells
- Causal or physical mechanisms
- Theoretical predictions or model comparisons

The narrowness of this scope is intentional. It keeps the measurement independent of interpretation and permits the data to be useful regardless of how later analyses proceed.

Derivative-Free Maximization: A critical design choice was the exclusion of $Z'(t)$ information from breadth certification. This decision prevents derivative approximation error from contaminating the breadth observable and avoids imposing stationarity constraints on a measurement that need not assume them. The 32-iteration golden-section search provides sufficient refinement for convergence to the reported tolerances.

Dual-Resolution Convergence: The independent measurement at two resolutions with fixed hard thresholds ($\epsilon_W < 10^{-9}$, $\epsilon_x < 10^{-5}$) ensures that refinement discrepancies remain negligible. The observed maxima of 2.63×10^{-15} and 8.96×10^{-8} indicate that even the most demanding cell achieved excellent numerical agreement. No cell was allowed to pass with marginal convergence.

Hypothesis Seal and Data Integrity: The explicit firewall against interpretation during acquisition protects against subtle forms of bias. The certified dataset reflects what was measured, not what was expected. This is especially important for datasets intended for later meta-analysis or reinterpretation under different theoretical frameworks.

5. Conclusion

This protocol provides a complete, reproducible specification for measuring the breadth and location of interior maxima of Hardy's Z-function in 10,000 consecutive zero-separated cells. All 10,000 cells passed full five-gate certification with refinement discrepancies far below preregistered limits. The

measurement is intentionally narrow in scope, protecting data integrity and enabling independent interpretation. The frozen specification, combined with the publicly available dataset, permits independent verification and replication of the measurement by other researchers. The data are available without restrictions to the scientific community.

References

- [1] Edwards, H. M. (1974). *Riemann's Zeta Function*. Academic Press.
- [2] Odlyzko, A. M. (1992). The 10^{20} -th zero of the Riemann zeta function and 70 million of its neighbors. In *Computational Methods in Number Theory*, Math. Centrum, Amsterdam.
- [3] Gourdon, P. (2004). Verification of the first trillion zeros of the Riemann zeta function. <http://numbers.computation.free.fr/Constants/Miscellaneous/zetacompute.html>
- [4] LMFDB Collaboration. (2024). The L-functions and Modular Forms Database. <http://www.lmfdb.org>

Appendix: Production Record Schema

The master CSV file contains the following 33 fields per cell (approximate logical structure):

batch_k, n, gamma_n, gamma_n1, h_n, A_points, A_dps, A_candidate_count, A_t_max, A_W_perp, A_x, B_points, B_dps, B_candidate_count, B_t_max, B_W_perp, B_rho, B_x, epsilon_W, epsilon_x, epsilon_W_limit, epsilon_x_limit, boundary_residual_left_50dps, boundary_residual_right_50dps, breadth_status, stationary_status, gate1_breadth_solver, gate2_interior_max, gate3_diagnostic_transparency, gate4_refinement_stability, gate5_raw_integrity, all_five_gates, notes

Field Descriptions:

Identification: batch_k (batch number 1–10), n (cell number 1–10000), gamma_n and gamma_n1 (consecutive zero ordinates)

Cell Geometry: h_n (cell width)

Resolution A: A_points (64), A_dps (28), A_candidate_count (number of local maxima found), A_t_max (location of selected maximum), A_W_perp (measured breadth)

Resolution B: B_points (128), B_dps (40), B_candidate_count, B_t_max, B_W_perp (primary breadth value), B_rho (breadth-to-width ratio), B_x (dimensionless location)

Convergence: epsilon_W (breadth discrepancy), epsilon_x (location discrepancy), epsilon_W_limit (10^{-9}), epsilon_x_limit (10^{-5})

Diagnostics: boundary_residual_left_50dps and boundary_residual_right_50dps (audit of zero boundaries at high precision)

Gates and Status: breadth_status, stationary_status, gate1_breadth_solver through gate5_raw_integrity (each: PASS or FAIL), all_five_gates (PASS only if all five gates pass)

Notes: Optional text field for flagging unusual cells or recording special circumstances